Price Mortality Theory: Full-Stack Train/Test Evidence with PMT, GARCH, and Combo Runs

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Abstract

This version rolls in *all* checked-in run artifacts: PMT grid sweep, AR(1)-GARCH(1,1), and the PMT+GARCH combination benchmark over the same S&P 500 snapshot (2019-01-01 to 2025-01-01). Train strength is still strong for PMT and strongest for the combo build (combo train median up to 0.509 by 42-day horizon). But out-of-sample is where the truth hits: PMT leads GARCH on test medians at 10–31 days, GARCH leads PMT on test means at every horizon and clearly at 42 days, and the current combo run underperforms both on test means at every horizon and turns negative by 31/42 days. Straight read: PMT still has signal, but it is horizon-sensitive; GARCH is more stable on cross-sectional means; and this combo implementation is not production-ready yet.

1 Introduction

Vol forecasting is still hard, and the market does not care about in-sample flexing. PMT is built to read structural trend decay from price alone through a multiscale surface, while GARCH is the baseline most desks already trust. So the only fair move is full train/test separation, same universe, same date range, then compare outputs cleanly.

Earlier drafts leaned on PMT's train wins. This rewrite keeps it real: all three model families are included, and conclusions are driven by out-of-sample behavior first.

2 Methods

Define the Simple Moving Average field $SMA(W,t)$ on closing price P_t. PMT "mortality" is:

$$\mu(W,t) = -\frac{1}{SMA}\left[\frac{\partial SMA}{\partial W} + \frac{\partial SMA}{\partial t}\right] \tag{1}$$

Finite differences approximate derivatives, and μ is lagged one period for causal evaluation.

Data and evaluation setup:

- Universe: frozen S&P 500 list in 'data/sp500$_tickers.txt$'$.Range: 2019-01-01to2025-01-01$.

- Split: chronological 70/30 train/test.
- PMT sweep: $W \in [20, 300]$ (step 20), $h \in [5, 60]$ (step 5), best (W^*, h^*) chosen on train.
- Horizon benchmark set: $h = \{10, 21, 31, 42\}$ for PMT-vs-GARCH and combo comparisons.

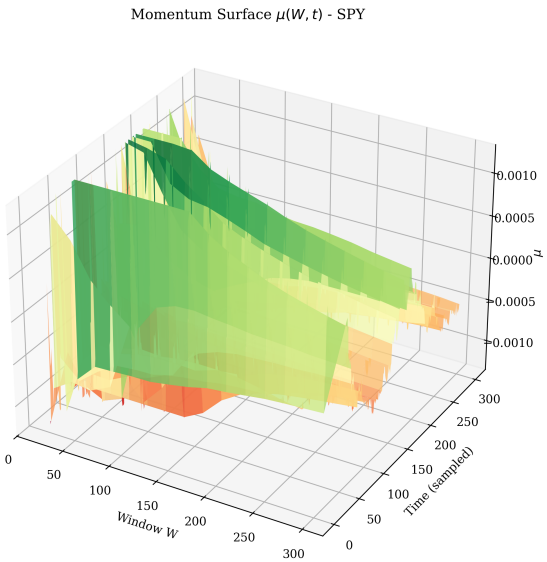

Figure 1: 3D momentum surface $\mu(W, t)$ for SPY (2019–2025).

3 Results

3.1 Coverage and run integrity

The latest checked-in manifests show 503 attempted tickers for each pipeline, with 497 PMT successes, 499 GARCH successes, and 500 combo successes. Failure logs are present for each run, so skipped names are auditable.

3.2 PMT full-grid (best-parameter) behavior

Across 497 PMT successes, in-sample quality remains strong (mean $\rho_{train} = 0.508$, median 0.531). Out-of-sample for those train-selected best parameters compresses sharply (mean $\rho_{test} = -0.018$, median 0.010; 489 non-null test observations).

Parameter selection remains concentrated at the edge of the search grid: W^* mean 241.4 (median 300) and h^* mean 43.0 (median 60). In plain terms: train optimization keeps pushing toward longer windows/horizons, but that preference does not transfer reliably to test.

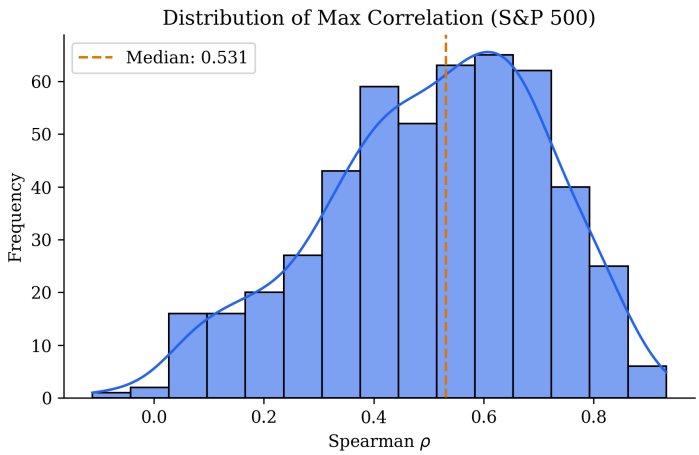

Figure 2: Distribution of PMT train-set max Spearman ρ across S&P 500 tickers.

3.3 Fixed-horizon PMT vs GARCH

At fixed horizons, PMT and GARCH tell a split story (Table **??**):

- PMT has stronger *test median* at 10, 21, and 31 days.
- GARCH has stronger *test mean* at every horizon.
- By 42 days, PMT mean/median are near zero to negative while GARCH remains positive.

Table 1: Out-of-sample PMT vs GARCH (Spearman ρ_{test} aggregates).

Horizon	PMT Mean	PMT Median	GARCH Mean	GARCH Median
10	0.074	0.087	0.087	0.051
21	0.069	0.063	0.079	0.033
31	0.033	0.048	0.066	0.016
42	-0.003	-0.002	0.059	0.010

Cross-sectionally, PMT beats GARCH on test ρ for about 47% of matched ticker-horizon pairs (not a majority), so this is not a clean takeover narrative.

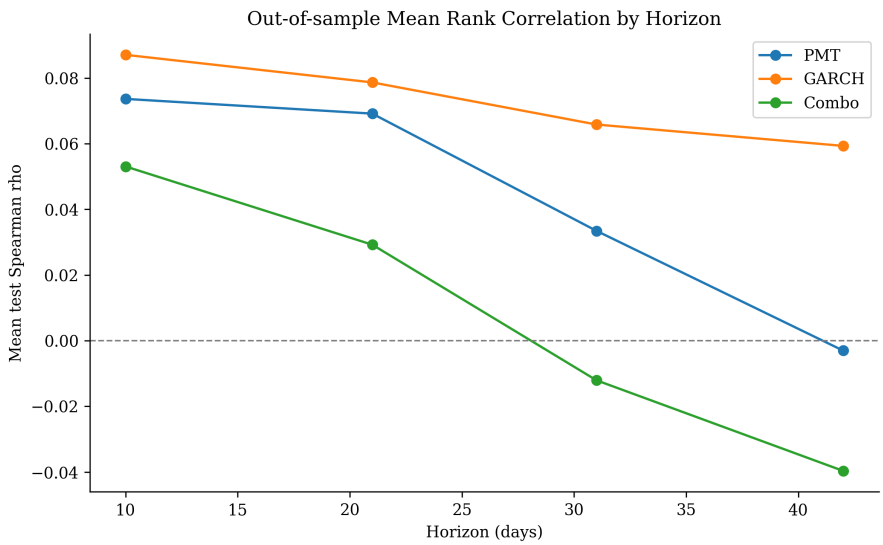

Figure 3: Out-of-sample horizon comparison using cross-sectional *mean* Spearman ρ_{test} for PMT, GARCH, and combo.

3.4 Combination run (PMT + GARCH)

The combo run was expected to improve robustness; right now it does not. It is very strong in-sample (train medians 0.463, 0.481, 0.498, 0.509 from 10 to 42 days), but test performance drops hard:

- Test means: 0.053 (10d), 0.029 (21d), -0.012 (31d), -0.040 (42d).
- Test medians: 0.059 (10d), 0.036 (21d), -0.012 (31d), -0.039 (42d).

Against PMT and GARCH in matched observations, combo wins only about 29.6% vs PMT and 43.8% vs GARCH on test ρ. So this blend, as implemented now, is overfit on train and not delivering out-of-sample lift.

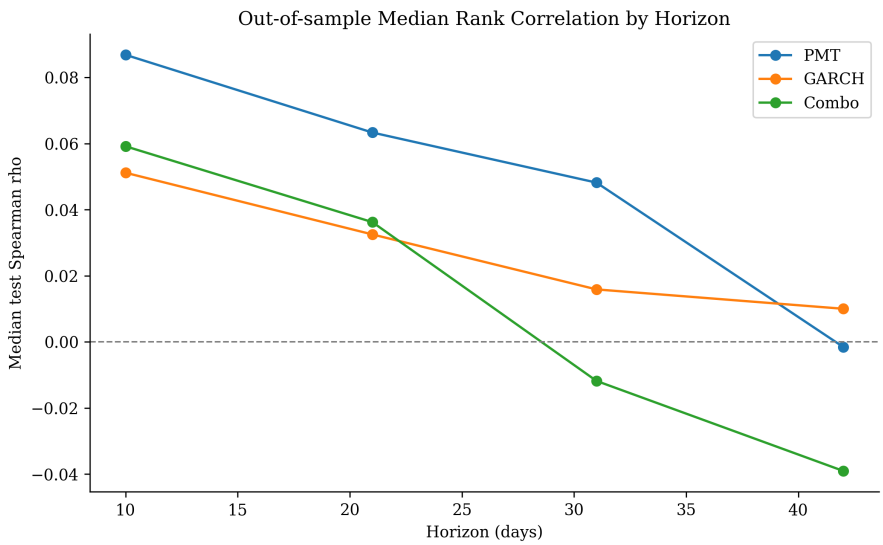

Figure 4: Out-of-sample horizon comparison using cross-sectional *median* Spearman ρ_{test} for PMT, GARCH, and combo.

4 Discussion

Professional summary, no hype:

- PMT is still useful, especially for short-to-mid horizons when judged by medians (cross-sectional robustness proxy).
- GARCH is the steadier choice on test means and becomes clearly better at longer horizons.
- The current combo stack is not ready; train gains are not surviving test.

So the deployment logic is simple: keep PMT in the feature set, keep GARCH as a backbone baseline, and treat combo architecture as an active research track rather than a shipped model. Next upgrades should prioritize regularization, stricter walk-forward validation, and horizon-specific weighting instead of static blending.

5 Conclusion

All-in evidence says PMT has real signal but not universal dominance. Horizon sensitivity is the key fact. GARCH remains competitive to superior on mean out-of-sample behavior, and the present PMT+GARCH combo run fails to add test alpha. The path forward is disciplined: horizon-aware calibration, anti-overfit controls, and ensemble design that is validated out-of-sample before any production claims.

Note: all metrics above are taken from checked-in CSV and manifest artifacts under 'results/'.

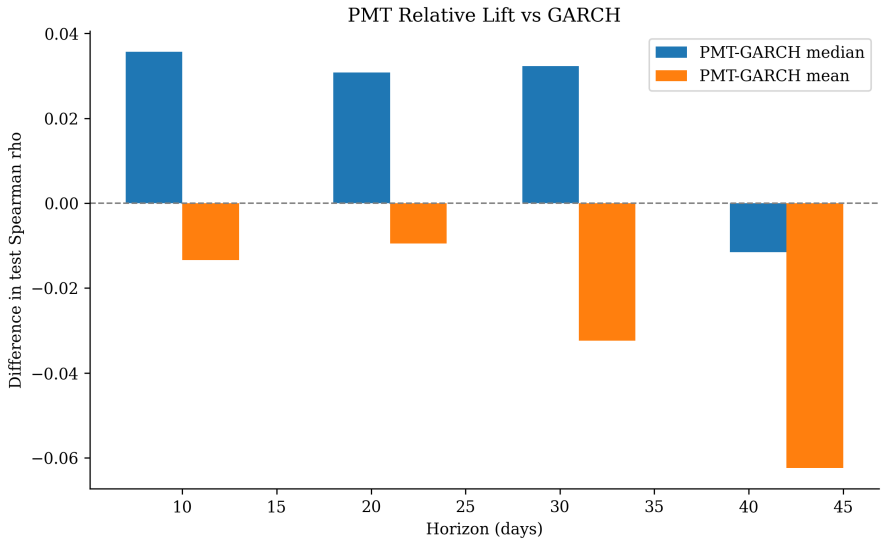

Figure 5: PMT minus GARCH out-of-sample ρ_{test} deltas across horizons (positive means PMT leads).

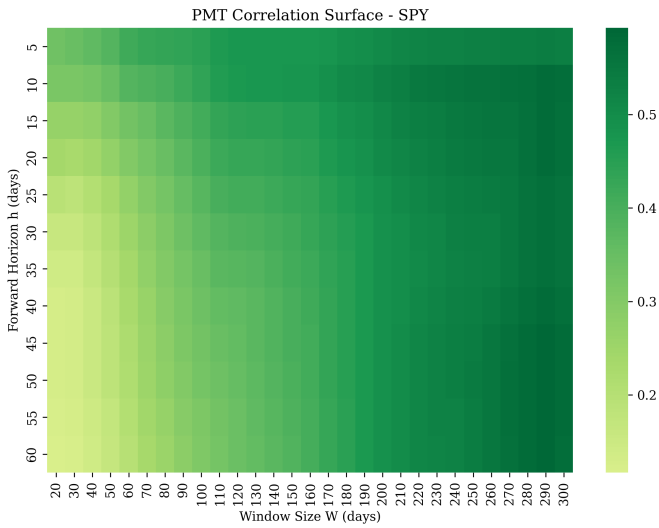

Figure 6: Spearman correlation surface for SPY (2019–2025).